

②

* Unit of potential : Joule/Coulomb = Volt

Unit of $\vec{E}$ -field : N/Coulomb = Volt-m

* Fundamental properties of $\vec{E}$ -field

① $\nabla \times \vec{E} = 0$ (Curl)

② $\oint \vec{E} \cdot d\vec{l} = 0$ for any closed path

line integral around a closed path

often $\vec{E} \cdot d\vec{r}$

③ $\nabla \cdot \vec{E} = \frac{\rho}{\epsilon_0}$

Maxwell-I

charge density (charge per unit volume)

$\oint \vec{E} \cdot d\vec{S} = \frac{Q}{\epsilon_0}$ charge enclosed

flux thru closed surface

Gauss's Theorem

Exercise: ① Show for $\vec{E} = \frac{1}{4\pi\epsilon_0} \frac{q_1 (\vec{r} - \vec{r}_1)}{|\vec{r} - \vec{r}_1|^3}$, by direct computation, that $\nabla \times \vec{E} = 0$

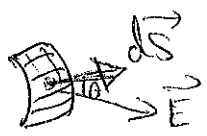
② Look up Stokes's theorem. Using $\nabla \times \vec{E} = 0$, use Stokes's theorem to prove $\oint \vec{E} \cdot d\vec{l} = 0$ for any

②

Gauss's law (Gauss's dielectric flux theorem)

Flux through an infinitesimal ~~patch~~ patch!

$$\vec{E} \cdot d\vec{S} = |d\vec{S}| \cos \theta$$



$d\vec{S}$ points perpendicular to surface & has magnitude equal to area of patch.

$\Phi = \int_{\Sigma} \vec{E} \cdot d\vec{S}$ definition of flux thru surface Σ

Gauss's law is about flux thru a closed surface:

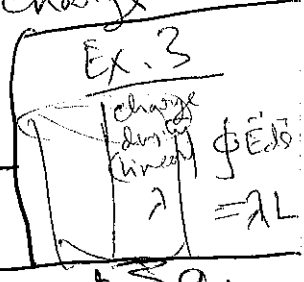
$$\oint_{\Sigma} \vec{E} \cdot d\vec{S} = \frac{Q_{enc}}{\epsilon_0}$$

Q_{enc} = total enclosed charge

Ex. 1

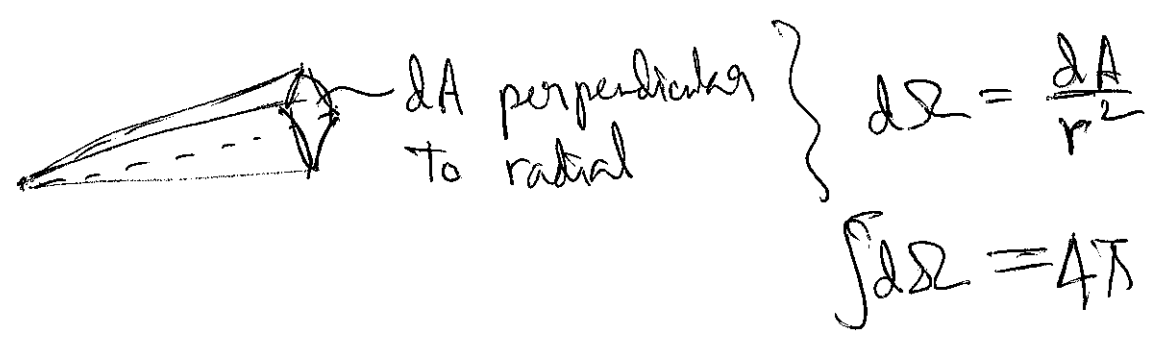


$$\oint \vec{E} \cdot d\vec{S} = \frac{q_1}{\epsilon_0}$$



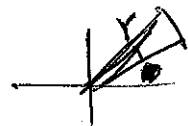
Ex. 2 $q_1, q_2, q_3, \dots$ $\oint \vec{E} \cdot d\vec{S} = \frac{\sum q_i}{\epsilon_0}$

To prove, we will use the concept of solid angle



(10)

Contrast to usual (planar) angle:

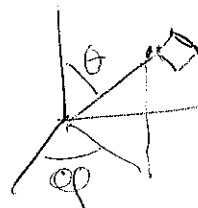


$$d\phi = \frac{dl}{r}$$

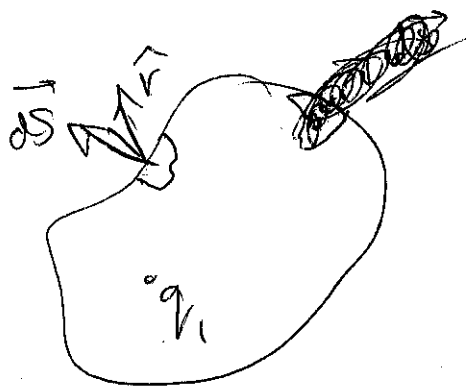
$$\int d\phi = 2\pi$$

If you know spherical coordinates,

$$d\Omega = \sin\theta d\theta d\phi$$



Proving Gauss's law for a single point charge enclosed!



$$\vec{E} \cdot d\vec{S}$$

$$= \frac{q_1}{4\pi\epsilon_0} \frac{\hat{r}}{r^2} \cdot d\vec{S}$$

$$\text{Now } \hat{r} \cdot d\vec{S} = |dS| \cos\theta = dA$$

$$\vec{E} \cdot d\vec{S} = \frac{q_1}{4\pi\epsilon_0} d\Omega$$

(perpendicular to radial direction)

$$\oint \vec{E} \cdot d\vec{S} = \frac{q_1}{4\pi\epsilon_0} \oint d\Omega = \frac{q_1}{\epsilon_0} \text{ for single point charge.}$$

* Using superposition principle! can extend to any collection/distribution of charges

* From Gauss's Theorem To Maxwell's 1st eq
and Poisson's Eq.

$$\oint_{\Sigma} \vec{E} \cdot d\vec{S} = \frac{Q}{\epsilon_0} = \frac{1}{\epsilon_0} \int_V \rho dV$$

Here V is the volume enclosed by Σ (Assuming continuous charge distⁿ)
Notation: $dV \equiv d^3r \equiv dx dy dz$

We use Gauss's divergence theorem! $\int_V \nabla \cdot \vec{A} dV = \oint_{\Sigma} \vec{A} \cdot d\vec{S}$

$$\oint_{\Sigma} \vec{A} \cdot d\vec{S} = \int_V (\nabla \cdot \vec{A}) dV$$

Exercise! Review
the divergence
thm and Stokes's
theorem.

$$\Rightarrow \int_V (\nabla \cdot \vec{E}) dV = \int_V \rho dV$$

This is true for arbitrary volume V .

$$\Rightarrow \boxed{\nabla \cdot \vec{E} = \frac{\rho}{\epsilon_0}} \rightarrow \text{Maxwell I}$$

Recalling $\vec{E} = -\nabla V$,

$$\boxed{\nabla^2 V = -\frac{\rho}{\epsilon_0}}$$

Poisson's equation.

(12)

* Applications of Gauss's law.

Application - I A

Infinite thin rod

Done in assignments using integration.

z

At $(x_0, 0, 0)$

$$\vec{E} = \frac{1}{2\pi\epsilon_0} \frac{\lambda}{x_0} \hat{x}$$

At a general point, using Cylindrical coordinates

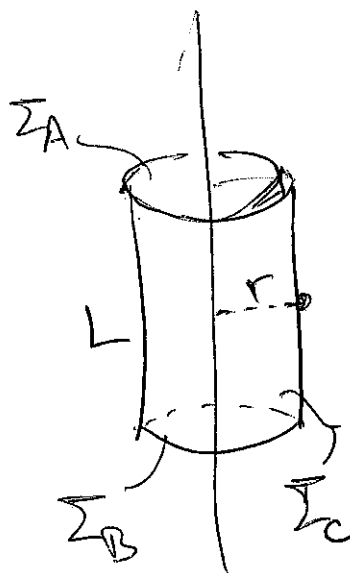
$$\vec{E} = \frac{1}{2\pi\epsilon_0} \frac{\lambda}{r} \hat{r}$$

Using Gauss's law

(Here r is the cylindrical coordinate.)

Symmetry:

Same $|\vec{E}|$, pointing outward, at all points of the curved part of cylinder.



$$\int_{\Sigma_C} \vec{E} \cdot d\vec{S} = |\vec{E}| \int_{\Sigma_C} |d\vec{S}|$$

$$= |\vec{E}| \times 2\pi r L$$

$$\Sigma = \Sigma_A \cup \Sigma_B \cup \Sigma_C$$

$$\int_{\Sigma_A} \vec{E} \cdot d\vec{S} = \int_{\Sigma_B} \vec{E} \cdot d\vec{S} = 0$$

$$\int_{\Sigma} \vec{E} \cdot d\vec{S} = 0 + 0 + |\vec{E}| 2\pi r L = \frac{2L}{\epsilon_0} \lambda$$

Gauss's
law

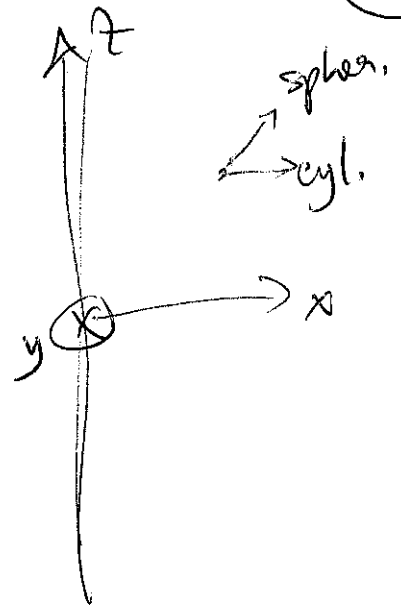
$$E = \frac{1}{2\pi\epsilon_0} \frac{\lambda}{r}$$

For thin charged rod,

$$\vec{E} = \frac{1}{2\pi\epsilon_0} \frac{\lambda}{r} \hat{r}$$

CAUTION

$\vec{r}, r, \hat{r}$



~~is a cylindrical coordinate/vector, not a spherical one.~~

~~is a cylindrical coordinate/vector, not a spherical one.~~

is a cylindrical coordinate/vector, not a spherical one.

* A CONDUCTOR : conducts charge.

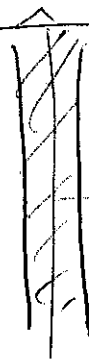
Static situation $\rightarrow$ all charge sits at surface.

Infinite thick rod of charge :

when metallic (conducting)



top view of charge density



When uniformly charged



top view of charge density

Could also consider HOLLOW rod



$\rightarrow$ For electric field OUTSIDE, internal structure doesn't matter

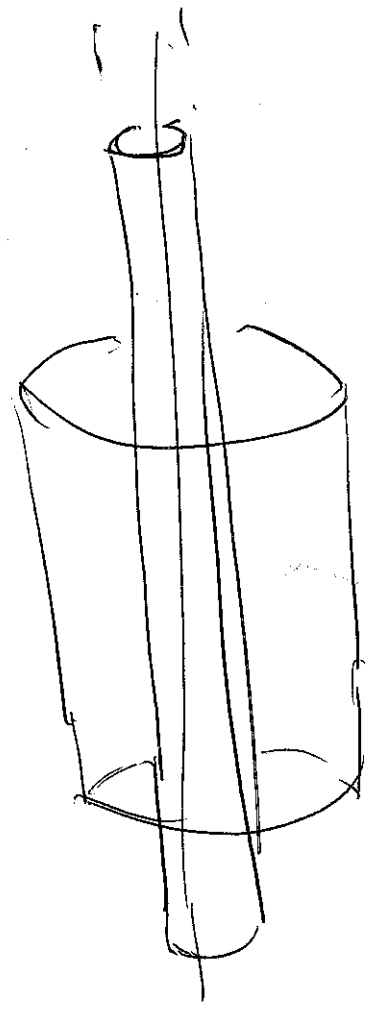
(14)

(15)

Applicatⁿ - IB Infinitely long cylinder —
~~infinitely long rod~~ either metallic
or filled with charge uniformly.

Charge λ per unit length.

charge resides
on outside

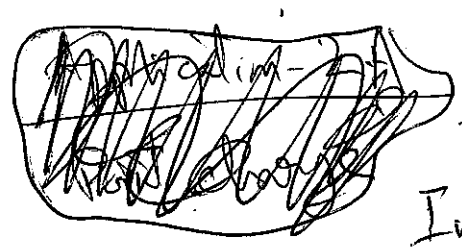


~~For~~ For electric field
outside the rod, exactly
the same argument holds.

$$\oint \vec{E} \cdot d\vec{S} = \frac{\lambda L}{\epsilon_0}$$

$$\Rightarrow E = \frac{1}{2\pi\epsilon_0} \frac{\lambda}{r}$$

As if it
was a thin
rod



Application 1C

Inside an infinite hollow charged
cylinder (or metallic cylinder)

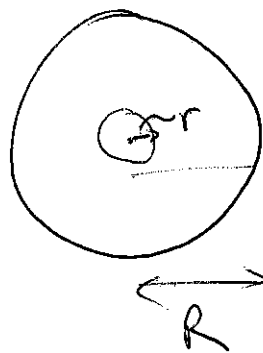
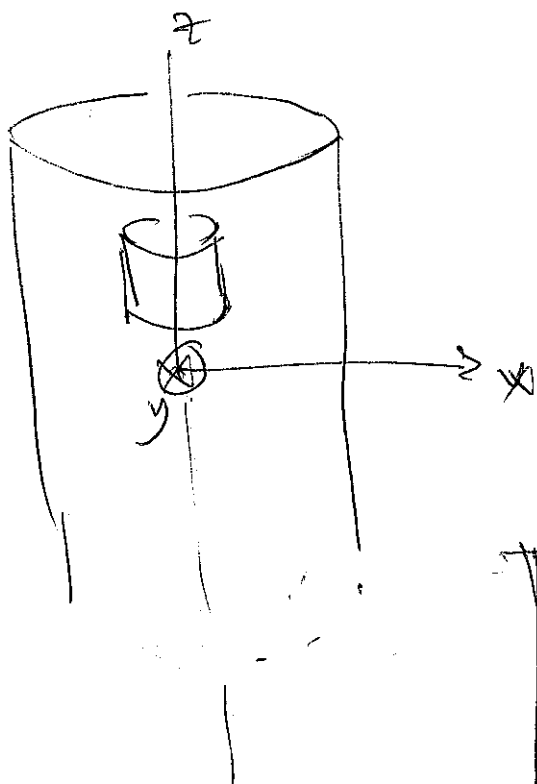


Exercise! Construct smaller
cylinder & show that $E=0$

Applicatⁿ 1D

Thick rod uniformly filled with charge.

CYLINDRICALLY symmetric. Charge density ρ .
Calculate $\vec{E}$ inside the rod:



Exercise: find the linear charge density of the rod, as function of R and ρ

$\vec{E}$ should depend on r only, and point radially ~~away~~ away from axis.

$$\oint_{\Sigma} \vec{E} \cdot d\vec{S} = \oint_{\text{curved}} \vec{E} \cdot d\vec{S} + \oint_{\text{caps}} \vec{E} \cdot d\vec{S}$$

$$= E \cdot 2\pi r L + 0$$

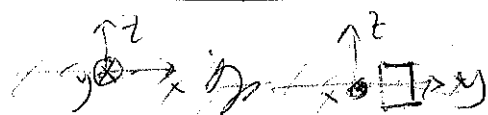
16

$$E \cdot 2\pi r L = \frac{Q}{\epsilon_0} = \frac{1}{\epsilon_0} \rho \pi r^2 L$$

$$\Rightarrow E = \frac{\rho}{2\epsilon_0} 2r$$

Exercise: Plot E as a function of r ,
~~clearly~~ clearly differentiating $r \leq R$,
 $r \geq R$ and $r > R$.

Application 2



Infinite sheet of charge (Problem Set 03)

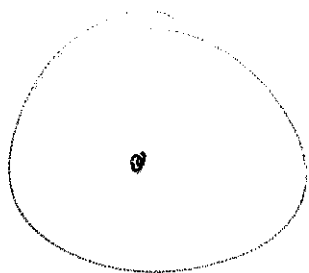


$$E \cdot A + E \cdot A = \frac{Q}{\epsilon_0}$$

$$\Rightarrow 2EA = \frac{\sigma A}{\epsilon_0} \Rightarrow E = \frac{\sigma}{2\epsilon_0}$$

Application 3a

Point charge Q



$$\oint \vec{E} \cdot d\vec{S} = E \cdot 4\pi r^2 = \frac{Q}{\epsilon_0}$$

$$E = \frac{1}{4\pi\epsilon_0} \frac{Q}{r^2}$$

→ "circular" derivation: we derived Gauss's ~~thm~~ from the eq., so of course Gauss's ~~thm~~ gives it back.